Prep for Texas Instruments

Who are we?

Texas Instruments Incorporated (TI) is a global semiconductor company that designs and manufactures semiconductors and various integrated circuits. TI is one of the top-10 semiconductor companies worldwide, based on sales volume and is focused on developing analog chips and embedded processors, which account for more than 80% of our revenue. TI has been a pioneer in many innovations in the semiconductor domain including the development of the first integrated circuit, the first patent on a single-chip microprocessor, the first single-chip linear predictive coding speech synthesizer, developing the prototype of the world's first transistor radio and the invention of the digital light processing device (also known as the DLP chip), which serves as the foundation for the award-winning DLP technology and DLP Cinema (used in IMAX theatres).

TI India was set up in 1985 and has R&D presence for all the major business units of TI including Analog (Data Converters, Amplifiers, Clocks & Synthesizers, Motor Drives, Power Management) and Embedded Processors (Connected Microcontrollers, Radar, ADAS - Advanced Driver Assistance, Infotainment Processors etc.) and caters to products for different market segments - Industrial, automotive, personal Electronics, Communication and Enterprise.

By employing the world's brightest minds, TI creates innovations that shape the future of technology. TI is helping about 100,000 customers transform the future, today. We're committed to building a better future – from the responsible manufacturing of our semiconductors, to caring for our employees, to giving back inside our communities and developing great minds. Put your talent to work with us – change the world, love your job!

To know more about TI, visit www.ti.com

Who are we looking for?

We are looking for individuals who are passionate about electronics, love to tackle challenging problems and build solutions that have an impact and can improve human lives though technology.

- **Basics, basics, basics!** Strong basics and fundamentals are what allow engineers to tackle new problems effectively
- **Perseverance** – people who quickly give up on a problem without trying or just saying "I don't know" without taking time to think, are pretty much immediately ruled out
- **Sharpness** – being able to identify patterns, and making logical leaps that can circumvent multiple steps
- **Thoroughness** – after making such a leap, being able to go back and justify the answer

How to prepare for the TI selection process?

Disclaimer: All the reference materials mentioned in this document have been collated based on suggestions/recommendations of individuals who have cleared the TI selection process. TI doesn't endorse these references. Users are advised to use their own discretion and judgement before utilizing these resources.

Analog

1. Basic Electrical Sciences
    a. Thevenin's Theorem, Norton's theorem, Superposition theorem
    b. Inputs can be step or sinusoidal in nature (Current sources or Voltage sources)
    c. Response to various inputs to circuits with combinations of R, L and C (very important)
    d. Second order RLC circuits and various types of responses (Overdamped, underdamped, critically damped, undamped)
    e. Should be able to intuitively explain a pole and a zero given a circuit and what are the major contributors for them
    f. Writing transfer functions of a RLC network and should be able to tell the "Q Factor", poles, zeros
    g. Two port network theory (knowing this is very handy while solving complex circuits)

    Reference material:

    a. Network Analysis – Van Valkenburg
    b. Circuit Theory – Hayt and Kemmerly

    Video lectures:

    - Lectures by Professor Nagendra Krishnapura (special attention to 2 to 11,18, 37, 38, 62 to 73, 108 to 138): https://www.youtube.com/playlist?list=PLa4KQhDlGd7QCTX3gTz0LyoL93jVjtaMe
    - Intro to Electrical Sciences lecture series (special attention to lectures 8, 9, 16, 34 to 49): https://www.youtube.com/playlist?list=PL6qRG5-NfbLvagdQOwShX9FMrzb5hSvrq
    - Some Lectures of IIT Madras: http://www.ee.iitm.ac.in/videolectures/doku.php?id=ec1010_2014nk:start

    Articles:

    Circuit Intuitions series: https://www.eecg.utoronto.ca/~ali/intuitions.html

    Charge sharing and capacitor basics:

    https://www.youtube.com/watch?v=NHeGgCeYSRM

2. Analog Electronics
    a. Positive feedback, Negative feedback, regulators, Oscillators, Barkhausen condition
    b. Everything from this course should be known from virtual ground concept till high pass – low pass filters
    c. Writing transfer function of opamp based configurations (which may include passive networks around the op amp)

    Reference material:

    a. Sergio Franco's Operational Amplifier book
    b. For quick revisions: http://www.electronics-tutorials.ws/
    c. Lectures by Professor Nagendra Krishnapura (special attention to 95 to 110): https://www.youtube.com/playlist?list=PLa4KQhDlGd7QCTX3gTz0LyoL93jVjtaMe

3. Signals & Systems
    a. Basics of LTI systems, Convolution theory, Nyquist theorem
    b. Frequency Domain – CT, DT, DTFT, DFT concepts should be clear and well understood

    Reference:

    - Alan V. Oppenheim 's book on Signals and systems, special attention to following parts:
        - Part 1: Signals and Systems
        - Part 2: LTI Systems
        - Part 3: Fourier Series
        - Part 4: CTFT
        - Part 5: DTFT
        - Part 7: Sampling
        - Part 9: Laplace Transform
        - Part 10: z-transform

    Video lectures:

    - Alan V. Oppenheim's lectures: https://www.youtube.com/playlist?list=PLADC1A1B7FA7FF7B6

4. Control Systems
    a. Bode plots, Stability criterion, etc
    b. Basic Feedback systems – What blocks are important in a Feedback System?
    c. Proper understanding of Second Order System Characteristics

d. Laplace & inverse Laplace transforms and estimating frequency response of output for step/impulse input

Reference material:

Books:

- Modern Control Theory – Kaushiko Ogata
- Control Systems Engineering – I.J. Nagrath and Kothari

  (Reference given is quite heavy with a lot of theory, important to concentrate on basics (that should be enough))

Video lectures:

- Analog Snippets videos on bode plots (special attention to #14 to 20): https://www.youtube.com/@analogsnippets/videos

5. Microelectronic Circuits
   a. Diode and Diode circuits
   b. Basic Functioning of a MOSFET – regions of operation
   c. Single Stage Amplifiers – CS, CG, CD Amplifiers gain, input & output impedance, with various loads, etc.
   d. Differential Amplifier
   e. Analysis using small signal circuit equivalent of all the basic amplifiers
   f. Miller's Theorem
   g. Basic Inverter circuit analysis

   Reference material:

   Books:

   - Microelectronics – Sedra and Smith, special attention to Part 1: Devices and Basic Circuits
   - Design of Analog CMOS Integrated Circuits – Behzad Razavi, special attention to following chapters:
     - Chapter 2: Basic MOS Device Physics
     - Chapter 3: Single Stage Amplifiers
     - Chapter 4: Differential Amplifiers
     - Chapter 5: Passive and Active Current Mirrors
     - Chapter 6: Frequency response of Amplifiers
     - Chapter 8: Feedback
     - Chapter 9: Op amps

– Chapter 10: Stability and Frequency Compensation

Video lectures:

- For diodes- Intro to Electrical Sciences lecture series (special attention to lectures 25 to 27 and 49):

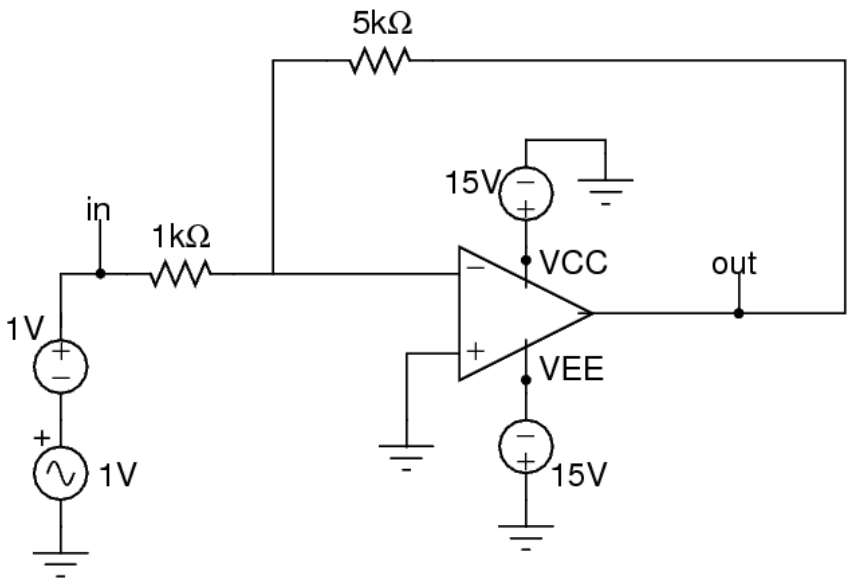
  https://www.youtube.com/playlist?list=PL6qRG5-NfbLvagdQOwShX9FMrzb5hSvrq

Useful Lectures by Dr. KRK Rao: http://nptel.ac.in/courses/117108107/

Other publicly available reference materials for Preparation:
https://www.youtube.com/@CircuitBread/videos
https://www.allaboutcircuits.com/

Sample Questions: *(For all the sample questions, try solving them before looking at the solutions. Remember, the approach is more important than the answer!* 😊*)*

1. You have designed the following op amp circuit such that it can take a 1V DC source and a 500-Hz sinusoidal source of amplitude 1V as input. What will be the output of the given op amp circuit?



Options:

i. 5 + 5 sin (2π.5000t)
ii. 5 -5 sin (2π.5000t)
iii. -5 + 5 sin (2π.5000t)
iv. -5 - 5 sin (2π.5000t)

Answer & explanation:

Correct Answer: -5 - 5 sin (2π.5000t)
There two sources - AC and DC at the inverting terminal. We'll consider them one at a time.
For DC source: By applying KVL and using virtual ground concept, we get Vo1 = -5 V
For AC source: By applying KVL and using virtual ground concept, we get = Vo2 = -Vs * 5 = -5sinwt = -5 sin (2 π 5000 t) therefore, Vo = Vo1 + Vo2 = -5 -5 sin (2 π 5000 t)

Here, KVL is Kirchhoff's Voltage Law

Points to Ponder:

- With limited gain of op-amp how the output voltage will change. (Higher, lower or same)
- If we change the input polarity, think of how the voltage output will look like for the same input.

2. You have connected a 200V, 50-Hz input voltage source to a load whose impedance is (2 + 2j) Ω. What will be the resultant power factor if you connect -4j Ω in parallel to the load?

Options:

i. 0.707 leading
ii. 0.707 lagging
iii. 0.5 leading
iv. None of these

Answer & explanation:

Correct Answer: None of these

Explanation: Calculating as mentioned in the question, 1/Z = (2+2j) || (-4j) = 4+0j

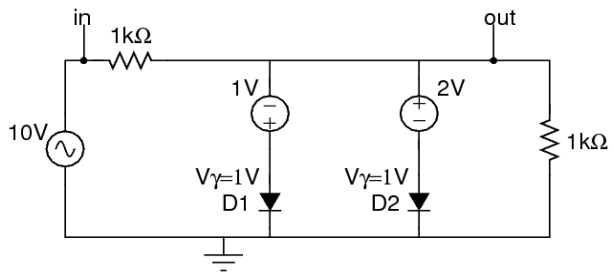
We know, power factor, cos Φ = R/Z = 4/4 =1

Therefore, Φ = 0

Points to Ponder:

- For maximum power transfer to happen between source and load what should be the impedance?

3. You are working on the following circuit wherein the cut-in voltage of both the diodes, **D1** and **D2** is **1V**. You now decide to apply a sinusoidal input voltage of frequency **1 kHz** and amplitude **10V.** What will happen the output voltage as a result of doing this?



Options:

i. It will be a sinusoidal with amplitude of 5V
ii. It will be a half-wave rectified sinusoid of amplitude 5V (only positive cycles)
iii. It will be a half-wave rectified sinusoid of amplitude 5V (only negative cycles)
iv. It will be a full-wave rectified sinusoid of amplitude 5V

Answer & Explanation:

Correct Answer: It will be a half-wave rectified sinusoid of amplitude 5V (only negative cycles)

Explanation: During positive half cycle, the resistances R1 and R2 will be in series. So, Vout = 5V (Using voltage divider rule). Now, for D1, the value at p-junction = 5+1 = 6V. For D2, the value of p-junction = 2v. Both D1 and D2 are forward biased. But, we'll consider the diode whose potential drop is maximum. So, D1 will be turned ON first. In that case, Vout = 0V, as diode will be replaced with 1V battery and they'll cancel each other. (+1-1= 0).

During negative half-cycle, both the diodes will be turned OFF, because potential at diode terminals will be negative. The path that will come into picture is the source and the two resistances. Hence, we get a half-wave rectified sinusoid of amplitude 5V (only negative cycles), because due to voltage divider, the output voltage will be 5V.

Points to Ponder:

- What will be output impedance in positive half cycle input and negative half cycle input if diode offer zero and infinite source impedance when on and off respectively?

General Interview Tips & Guidelines

CV:

1. Clearly mention key achievements, projects, papers / publications, interest areas and grades in relevant courses

Personal Interview:

2. Think out loud while solving any question. The Interviewer judges you on the approach and not the answer

3. Pay attention to the question and think before proceeding, it is okay to take few seconds before answering if you are not sure
4. If you don't know the answer or are not sure about the answer, then mention the same before answering
5. There might be questions on areas that you have not worked on. That does not mean you completely give up and not answer. Try answering the questions based on the hints provided by the interviewer. The interviewer is trying to see if you would be able to solve a question logically even though you do not have a background on the topic
6. Show curiosity and your drive to solve problems. Ask questions, seek help and think aloud
7. If you have hit a roadblock, don't be afraid to go back to square one and start afresh on a new path